SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME:Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

*CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)*

Duration: 1 Hour
QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

**Introduction**

A Neural Network is a supervised machine learning model used to predict continuous values such as student scores. The performance of a regression model is evaluated using **Mean Squared Error (MSE)**, which measures the average squared difference between the actual and predicted values. A lower MSE indicates better prediction accuracy. In this problem, the training MSE is **0.85**, whereas the testing MSE is **5.20**. The large difference between these two values shows that the model performs well on the training dataset but poorly on unseen testing data. This indicates that the model is unable to generalize effectively.

**Step 1: Error Identification**

**Given:**

- Training MSE = 0.85
- Testing MSE = 5.20

The testing error is much higher than the training error. This clearly indicates that the neural network is suffering from **overfitting**. Overfitting occurs when the model memorizes the training data instead of learning the general relationship between the input features and the target values. As a result, it achieves excellent accuracy on the training dataset but fails to make accurate predictions on new data.

**Step 2: Error Analysis and Reasoning**

The primary reason for the high testing error is overfitting. During training, the neural network has learned not only the useful patterns but also the noise and unnecessary details present in the training dataset. Consequently, it performs extremely well on the training data but cannot generalize to unseen data.

Another possible reason is that the neural network is too complex. If the model contains many hidden layers and a large number of neurons, it can easily memorize the training samples instead of learning meaningful relationships. Such a complex model usually requires a very large amount of training data.

A limited training dataset may also contribute to the problem. When only a small number of student records are available, the model does not observe enough variations in student performance. Therefore, it becomes difficult for the network to make accurate predictions for new students.

The presence of noisy or incorrect data is another important factor. Missing values, incorrect marks, duplicate records, or wrongly labeled samples can mislead the learning process. Instead of identifying genuine academic patterns, the neural network learns incorrect relationships.

The absence of regularization techniques such as Dropout or L2 Regularization further increases the possibility of overfitting. Without regularization, the model becomes highly dependent on specific neurons and loses its ability to generalize.

Poor feature selection is another possible cause. Including irrelevant features such as Student ID, Roll Number, or Registration Number increases the complexity of the model without improving prediction accuracy. These unnecessary features reduce the efficiency of the learning process.

Differences between the training and testing datasets may also produce high testing error. For example, if the training data contains students from one institution while the testing data contains students from different institutions, the academic performance patterns may vary considerably, resulting in poor predictions.

Improper hyperparameter selection, including an unsuitable learning rate, excessive number of epochs, incorrect batch size, or inappropriate activation functions, can also negatively affect the performance of the neural network.

**Step 3: Correction Strategy**

Several corrective measures can be implemented to improve the model's performance. The first and most important technique is to apply **L1 or L2 Regularization**, which penalizes large weight values and reduces model complexity. This helps prevent overfitting and improves generalization.

Another effective technique is **Dropout**. During training, dropout randomly deactivates a percentage of neurons, forcing the remaining neurons to learn more robust and independent features. This significantly reduces overfitting.

**Early Stopping** should also be applied. During training, the validation error is continuously monitored. Once the validation error starts increasing while the training error continues to decrease, the training process should be stopped. This prevents the model from memorizing the training data.

Increasing the size of the training dataset is another important solution. More student records from different academic years or institutions enable the neural network to learn a wider variety of patterns and improve prediction accuracy.

Data preprocessing should also be performed carefully. Missing values should be handled appropriately, duplicate records should be removed, noisy samples should be corrected, and numerical features should be normalized. Clean data enables the neural network to learn more meaningful relationships.

Feature engineering can further improve the model. Useful features such as attendance percentage, study hours, previous semester GPA, assignment scores, and internal assessment marks should be retained, while irrelevant attributes should be removed.

Cross-validation techniques such as **K-Fold Cross Validation** should be used to evaluate the model on multiple subsets of the dataset. This provides a more reliable estimate of model performance and reduces dependence on a single train-test split.

Reducing the complexity of the neural network by decreasing the number of hidden layers or neurons is another effective strategy. A simpler model often performs better when the available dataset is relatively small.

Finally, hyperparameter tuning should be carried out to determine the optimal learning rate, batch size, optimizer, activation functions, and number of epochs. Techniques such as Grid Search or Random Search can be used to identify the best combination of parameters.

**Step 4: Interpretation of Results**

Initially, the model achieved a **Training MSE of 0.85** and a **Testing MSE of 5.20**. The large difference between these values clearly indicates poor generalization and severe overfitting. Although the model appears highly accurate during training, it cannot produce reliable predictions on unseen student data.

After implementing the suggested corrective measures, the testing error is expected to decrease significantly while the training error may increase slightly. For example, the training MSE may become **1.10** and the testing MSE may reduce to **1.35**. The small difference between these values indicates that the neural network has successfully learned the underlying relationship between student characteristics and academic performance rather than memorizing the training data. As a result, the model becomes more reliable, stable, and accurate when predicting student scores for new datasets.

**Conclusion**

The neural network exhibits **overfitting** because the **Training MSE (0.85)** is much lower than the **Testing MSE (5.20)**. The main causes include excessive model complexity, insufficient training data, noisy datasets, lack of regularization, poor feature selection, differences in data distribution, and improper hyperparameter settings. By applying techniques such as **Regularization, Dropout, Early Stopping, Cross-Validation, Data Cleaning, Feature Engineering, Increasing Training Data, Hyperparameter Tuning, and Model Simplification**, the testing error can be reduced significantly. These improvements enable the neural network to generalize effectively and produce accurate predictions for unseen student data, thereby enhancing the overall performance and reliability of the model.

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*Code of Conduct:*

*I **T Bhupal/192425243** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:* ***bhupalreddy***

RUBRICS FOR CALCULATION

| **Criteria** | **Weightage** | **Level 4 – Exemplary (5)** | **Level 3 – Proficient (4)** | **Level 2 – Developing (2-3)** | **Level 1 – Beginning (0-1)** |
|---|---|---|---|---|---|
| Error Identification | 30% | Correctly identifies all errors and their causes | Identifies most errors with minor omissions | Identifies some errors but misses key issues | Unable to identify errors |
| Error Analysis & Reasoning | 30% | Provides clear and logical explanation of why errors occurred | Reasoning mostly corrects with minor gaps | Limited reasoning and partial explanation | No meaningful analysis |
| Correction Strategy | 25% | Suggests appropriate corrections with justification | Suggests correct corrections with minor gaps | Partially correct corrections | Incorrect or no corrections |
| Interpretation of Results | 15% | Clearly explains impact of errors on model performance/results | Basic interpretation with minor omissions | Limited interpretation | No interpretation provided |